CH
07

Build Tools

Maven Multi-Module · Gradle · Plugins · CI Integration

Overview

Maven multi-module primary build; Gradle side-by-side demo. Plugin-enforced quality gates, layered Docker images, and CI integration.

Maven Multi-Module Structure

- **shopverse-parent (BOM + multi-module root, <packaging>pom</**
  - shopverse-domain – zero framework deps; pure Java recor
  - shopverse-application – use cases, ports, command/query hand
  - shopverse-infrastructure – JPA, Redis, Kafka, MongoDB, Cassa
  - shopverse-web – controllers, DTOs, Spring MVC, secur
  - shopverse-gateway – Spring Cloud Gateway
  - shopverse-e2e-tests – Testcontainers + Pact contract tests
  - benchmarks/ – JMH benchmarks module

Key Maven Plugins

Plugin	Goal / Config	Purpose
spring-boot-maven-plugin	repackage + buildInfo + layered JAR	Executable JAR; /actuator/info; Docker layer caching
flyway-maven-plugin	mvn flyway:migrate	Run migrations in CI before integration tests
jacoco-maven-plugin	haltOnFailure=true, min 80% line coverage	Enforced code coverage gate
maven-enforcer-plugin	Java 21, Maven 3.9+, no SNAPSHOT in prod profile	Dependency governance
spotbugs-maven-plugin	Fail on HIGH severity	Static analysis

Maven vs Gradle Comparison

Dimension	Maven	Gradle
Config language	XML (pom.xml)	Kotlin/Groovy DSL (build.gradle.kts)
Build performance	Good (incremental via plugins)	Faster (incremental + build cache)
Dependency management	BOM / <dependencyManagement>	Version catalog (libs.versions.toml)
Plugin ecosystem	Very large (Maven Central)	Large (Gradle Plugin Portal)
Learning curve	Low (convention-driven)	Medium (flexible = complex)
IDE support	Excellent (all IDEs)	Excellent (all IDEs)
ShopVerse choice	Primary build tool	Side-by-side demo in shopverse-infrastructure

Dependency Scopes

Scope	Classpath	Packaged	Example
compile (default)	compile + runtime	Yes	spring-boot-starter-web
test	test only	No	testcontainers, junit
provided	compile only	No	lombok
runtime	runtime only	Yes	postgresql JDBC driver